



Rensselaer

Course Syllabus

Course Information

Digital Electronics

RPI Spring 2026

3 credits

ECSE-4040

Section 1

Lecture	Section 1	TF	10:00PM–11:20PM	JEC 4104
Studio	Section 1	W	12:00PM–2:00PM	JEC 4309

Course Website: lms.rpi.edu

Prerequisites or Other Requirements:

ECSE-2050 *Introduction to Electronics*; ECSE-2610 *Computer Components and Operations*.

Taught Spring Term, Annually.

6 Contact Hours.

Instructor

T. Paul Chow

Office Location: CII-6111

Office Hour: T 1:00–2:00PM

chowt@rpi.edu

(518) 276-2910

Teaching Assistant(s)

Name	Office	Office Hours	Email Address
Zhaowen He	CII-6117	F 10:00-11:00AM	hez5@rpi.edu

Course Description

Analysis and design of switching-mode circuits: NMOS, CMOS, RTL, DTL, TTL, and ECL digital-logic families. Topics include the following: basic logic gates (voltage-transfer characteristics, noise margin, fan out, propagation delay, power dissipation), flip flops, Schmitt triggers, oscillators, timers, memories, A/D and D/A converters, and optional advanced topics.

Course Text(s)

1. D.A. Neamen, *Microelectronics: Circuit Analysis and Design*, 4th Ed., McGraw-Hill, 2010.
2. *LTspice* Freeware (see www.linear.com/designtools/software).

Supplemental References

1. D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.
2. N.H.E. Weste and D.M. Harris, *CMOS VLSI Design: A Circuits and Systems Perspective*, Addison-Wesley, 2011.

3. J.M. Rabaey, A. Chandrakasan, and B. Nikolic, *Digital Integrated Circuits: A Design Perspective*, 2nd Ed., Prentice-Hall, 2003.
4. A. Vladimirescu, *The SPICE BOOK*.
5. R.F. Pierret, *Semiconductor Device Fundamentals*, Pearson, 1996.

Course Goals / Objectives

We like to teach students how to analyze and understand basic digital integrated circuits (ICs). The ability to perform both 'Hand' circuit calculations and Spice circuit simulations is stressed. Particular emphasis is placed on MOS and BJT switching circuits. We treat, in some depth, advanced digital circuits---including regenerative circuits, IC memories, and digital converters. We expect, of note, that students will become familiar with the meaning of common terms and acronyms, and critical issues associated with modern digital IC technology.

Course Content

1. Introduction.
2. MOSFETs.
3. MOS Inverters.
4. MOS Gates.
5. Ebers-Moll Model & BJT Transients.
6. BJT Inverters.
7. BJT Gates.
8. Regenerative Circuits.
9. Memories.
10. A/D & D/A Converters.

Student Learning Outcomes

1. ... skills for hand circuit analysis of basic MOS digital gates: voltage-transfer characteristic (VTC), propagation delay time.
2. ... skills for hand circuit analysis of basic BJT digital gates: voltage-transfer characteristic (VTC), fan out, average power dissipation, propagation delay time.
3. ... an ability to verify the results of hand MOS digital-circuit calculations by constructing appropriate network topologies and simulating them in Spice.
4. ... an ability to verify the results of hand BJT digital-circuit calculations by constructing appropriate network topologies and simulating them in Spice.

Course Assessment Measures

Assessment	Due Date	Learning Outcome Nos.
Hand/Spice Problems	Nearly Once Per Week	1, 2, 3, 4
Exam	Once, Mid-Semester	1, 3
Exam	Once, Finals Week	1, 2, 3, 4

Grading Criteria

Studio Problems: 30%
Mid-Term Exam: 30%

Final Exam: 40%

Note: Studio Percentile Score = (Total Number of Studio Points / 240) x 35.

Attendance Policy

Spot attendance may be taken during Lectures. This data may be used to determine a final course grade, should you fall onto a borderline. Note: Attendance and Participation during Studio is absolutely mandatory! We will not accept any completed Studio Problems without your required Wednesday Studio attendance during any particular week! (Should you attempt to submit, without having attendance before the Due Date, we will not accept your work!)

In this regard, make sure to see the Instructor, immediately, if possible, BEFORE missing a future Lecture or Studio. Excuses related to valid personal emergencies will always be accepted; however, the Instructor will judge acceptance of non-emergency excuses. Multiple non-emergency excuses and those beyond one week of the 'incident' will, generally, not be accepted.

Other Course Policies

GRADING:

Studio Problems will generally be graded on the basis of evidence and quality of work. The grading scale is usually “digital”: 1 point per HAND and per SPICE problem, when completed by the specified due date; otherwise, 0 points per HAND and per SPICE problem. Mid-Term Exam and Final Exam will be graded primarily on the basis of correctness of answer, with partial credit possible on the basis of evidence of work. The grading scale for each Exam is 'analog': 0-100 points.

Final grades in the course will be evaluated on the basis of a relative scale (or “curve”) and ascribed on a “non-fractional” basis (“A”, “B”, “C”, “D”, “F”). Best estimates of your current standing, that is, projected grade, will be furnished when possible.

The Course Instructor reserves the right to make final judgments on re-grading! The re-grading appeal process is as follows: (1) see the Teaching Assistant; if still not satisfied, (2) see the Course Instructor; and, if necessary, (3) the Instructor will direct you to any higher-level university authorities with whom you may wish to speak. Note: during the re-grading process, the Course Instructor reserves the right to re-grade the entire exam, not just the portion in question. All exams are mandatory! Valid conflicts must be settled with the Instructor before---not after---the exam date!

COURSEWORK & LOGISTICS:

The Instructor will assign problems for each Studio session. Your grading TA will look over and grade your work to see that a proper effort was generated. Students are encouraged to work in small (<5-person) groups. The top Studio Problem submission page must include Student Name, Student ID, Date; individual problems must be marked with their numerical label. Though submission is electronic (via LMS), please keep a stapled hard copy of work should we need to refer to it later during the semester.

Neither the Course Instructor nor the Course Secretary will accept Studio problem assignments (late, or otherwise)! You must seek Instructor approval, after Lecture regarding late work. Seek this approval before the due date, in non-emergency cases. It is then your responsibility to personally direct late Studio problems to the appropriate (grading) TA with an accompanying note of explanation (include it in your LMS submission text box). The TAs will, at their convenience, mark the day of receipt and determine the degree of credit (full, partial, none) after checking with the Instructor. No exceptions!

The Course Instructor reserves the right to change Studio problems, alter Due Dates, or otherwise make exceptions the rules given above.

Both Mid-Term and Final Exams. have in-class and take-home portions. The in-class portion typically lasts 30-40 minutes and happens at the beginning of the Studio session on the Wednesday that the take-home portions are due.

The take-home portion of the Mid-Term and Final Exam is distributed, submitted and returned via LMS. No credit will be given for late return, after its Due Date. Please contact the Instructor before the Exam if you have any concerns about its timely completion. Contact the Instructor as soon as possible should you encounter any personal emergencies in connection with these Exams.

It should be *emphasized* that no generative AI tool (such as ChatGPT) of any kind is allowed in answering of Studio Problems and Exams. Should it be discovered that a student utilizes such tools, that entire Studio Problem or Exam will be given an F grade/0 point.

RETURN OF GRADED MATERIAL:

Your grading TA will score your Studio Problems according to previously described guidelines. Your graded Studio Problems and the Exams. will be returned electronically, via LMS, with any necessary text commentary. It is your responsibility to safely store your hard copy coursework for future reference or grade verification, should it be necessary.

PROBLEM SOLUTIONS:

Unlike most other courses, generation of Studio Problem solutions is the responsibility of each student. You are asked to develop your solutions in Studio. In Studio, solutions will be revealed to check your work as you progress. Attending Studio, therefore, is important. Please make sure you ask sufficient questions in Studio, as well. (Note, it is expected, in Studio, that you have attended Lecture that week. Missing Lecture can seriously impact your performance, and the help TAs can effectively provide during Studio. Please make every effort to attend Lecture!)

Instructor Solution Sketches will be provided on LMS after the Studio Problem Due Date. Generally, the sketches cover the HAND portion of your assignments; the SPICE portion should have been worked out during Studio each week. Any questions in this regard should be sent to your TA(s). They will go over any of your corrected work, as needed.

Academic Integrity

Student-teacher relationships are built on trust. For example, students must trust that teachers have made appropriate decisions about the structure and content of the courses they teach, and teachers must trust that the assignments that students turn in are their own. Acts that violate this trust undermine the educational process. The Rensselaer Handbook of Student Rights and Responsibilities defines various forms of Academic Dishonesty and you should make yourself familiar with these. In this class, all assignments that are turned in for a grade must represent the student's own work. In cases where help was received, or teamwork was allowed, a notation on the assignment should indicate your collaboration.

Violation of the Academic Integrity Policy described here will result in a penalty. The penalty will typically involve an assignment or exam grade reduction or a course grade reduction, along with a referral to the Senior Judicial Administrator in the Dean of Students Office.

If you have any question concerning this policy before submitting an assignment, please ask for clarification.

Excuses given for late assignments or conflicts with Exams should be truthful. The best chance of having an excuse accepted is the truth. The truth will draw the most “sympathy” from the Course Instructor as honesty has a high value here at Rensselaer. Note, importantly: it is required, in non-emergency situations (that is, not involving personal or family health issues), that you personally furnish your excuse to the Course instructor after Lecture, and before the required normal due date.

PERMITTED ACADEMIC ACTIVITIES:

You are permitted to work in small groups to complete the Studio Problem assignments. (This includes HAND and SPICE assignments.) Students are, however, only permitted to consult with each other in their groups. Copying, paraphrasing, or having someone else do these assignments is not permitted, whether or not the student(s) involved is (are) in a group. No grading penalty will be ascribed to those who work in groups, relative to those who do not. Note, the use (copying, paraphrasing, etc.) from a Solution Sketch or Manual, another student's solutions, including those from previous years (hard copy, electronic copy, computer printouts, and so forth.) is strictly prohibited. No use of someone else's work (present or previous years), nor use of unofficial aids (consulting with anyone but the Course Instructor) is permitted during the Mid-Term Exam. No use of someone else's work, nor use of unofficial aids (“cheat sheets”; storing and/or using information from calculators, mobile devices or computers; student imposters; writing on the hand or other object; sign or sound language, and so forth) is permitted during the Final Exam.

The use of generative AI tools to complete any aspect of assignments or take-home exams. for this course are not permitted and will be treated as plagiarism. If you have questions about what constitutes a violation of this statement, please contact the Instructor (see <https://info.rpi.edu/teaching-and-learning-collaboratory>).

Diversity, Equity and Inclusion

The Instructor firmly believes that diversity in classroom will foster a conducive environment for education and learning. Hence, he strives to create an open and honest space for students to discuss and exchange different ideas and approaches in problem solving, while he only tolerates the

highest level of respect for each other. See the DEI statement of RPI (see <https://info.rpi.edu/diversity>).

Other Course-Specific Information

EXAM COVERAGE:

Mid-Term: MOS Gates, Course Notes, Chaps. 1-3.

Final: MOS and BJT Gates, Memory Circuits, Course Notes, Chaps. 4-7.

IMPORTANT DATES:

Mid-Term Exam Start: W, Feb. 11, after Studio.

Mid-Term Exam Due: W, March 11, beginning of Studio.

Final-Exam Review: Tu, April 28. (All during Lecture.)

Final Exam: May 4–8 (Day, Location TBA).

COURSE SECRETARY:

Mr. Jody AMundson

Email: amundj@rpi.edu

Tel: x-6724, Rm: CII-6015, Office Hrs: M-F, 9am-4:00pm

Course Calendar

Week	Date	Topic	Readings	Assignments
Wk 1	1/12	Lecture: Tu, Introduction; F, MOSFETs.	Neamen: pp. 1141 – 1144; LTspice Web Site (www.linear.com/ltspice).	Studio: W, No Class.
Wk 2	1/19	Lecture: Tu, MOSFETs; F, MOSFETs.	Neamen: Ch 3. pp. 125 – 146; HJS 3 rd Ed: Ch. 2 pp. 35 – 64.	Studio: W, LT SPICE Tutorial; P2.1, 2.5 (HJS 3 rd Ed.)
Wk 3	1/26	Lecture: Tu, MOSFETs (<i>Parameters, Models</i>); F, MOS Gates (<i>VTC, Noise Margins</i>).	Neamen: Ch. 4 pp. 205 – 215; HJS 3 rd Ed: Ch. 3 pp. 111 – 138.	Studio: W, Parameter extraction, P3.5 (HJS 3 rd Ed.).
Wk 4	2/2	Lecture: Tu, MOS Gates (<i>Inverters</i>); F, MOS Gates (<i>Sizing, VTC, NAND, NOR</i>).	Neamen: Ch. 16 pp. 1147 – 1202; HJS 3 rd Ed: Ch. 4 pp 142 – 180, Ch. 5 pp. 195 – 205.	Studio: W, P4.1 & 4.2 (HJS 3 rd Ed.).
Wk 5	2/9	Lecture: Tu, MOS Gates(<i>RC Delays, Combinational Logic</i>); F, MOS Gates(<i>Noise Sources, NM-Delay, Sequential Logic</i>).	HJS 3 rd Ed: Ch. 4 pp 181 – 186, Ch. 5 pp. 206 – 227.	Studio: W, P4.10 & 4.13 (HJS 3 rd Ed.). <u>MID-TERM EXAM STARTS</u> (at the end of Studio).
Wk 6	2/16	Lecture: Tu, No Class ; F, MOS Gates (<i>Sequential Logic, Power</i>).	HJS 3 rd Ed: Ch. 5 pp. 228 – 242.	Studio: W, P4.14, 5.1, & 5.7 (HJS 3 rd Ed.).
Wk 7	2/23	Lecture: Tu, MOS Gates (<i>Power</i>); F, MOS Gates (<i>Propagation Delay, Capacitance</i>).	HJS 3 rd Ed: Ch. 2 pp. 79 – 83, Ch. 6 pp. 249 – 267.	Studio: W, P5.12 & 5.16 (HJS 3 rd Ed.).
	3/2	Spring Break		
Wk 8	3/9	Lecture: Tu, MOS Gates (<i>Input Slope, Elmore Delay</i>); F, Interconnects.	HJS 3 rd Ed: Ch. 6 pp. 267 – 277; Ch. 3 pp. 100 – 103, Ch. 10 pp. 441 – 468.	Studio: W, P6.4 (HJS 3 rd Ed), Check Studio Grades. <u>MID-TERM EXAM DUE</u> (at beginning of Studio).
Wk 9	3/16	Lecture: Tu, BJT (<i>Ebers-Moll Model</i>); F, BJT Inverter (<i>VTC</i>).	Neamen: Ch. 5 pp. 285 – 326; HJ 2 nd Ed: Ch. 5 pp. 108 – 124, Ch. 6 pp. 126 – 147.	Studio: W, P5.1, 5.2, & P6.1 – 6.4 (HJ 2 nd Ed.).
Wk 10	3/23	Lecture: Tu, BJT Gates (<i>ECL</i>); BJT Gates (<i>Low Power ECL</i>).	Neamen: Ch. 17 pp. 1256 – 1276; HJ 2 nd Ed: Ch. 7 pp. 173 – 179, Ch. 8 pp. 192 – 194.	Studio: W, No Lab due to GM Week.

Week	Date	Topic	Readings	Assignments
Wk 11	3/30	Lecture: Tu, BJT Gates (<i>CML</i>); F, Regenerative Ckts (<i>Mono & Astable</i>).	HJ 2 nd Ed: Ch. 8 pp. 202 – 209.	Studio: W, P7.9– 7.12 (HJ 2 nd Ed.).
Wk 12	4/6	Lecture: Tu, Memories (<i>SRAM</i>); F, Memories (<i>DRAM, Sense Amplifiers</i>)	Neamen: Ch. 16 pp. 1208 – 1221; HJS 3 rd Ed: Ch. 8 pp. 368 – 377; WH 4 th Ed: Ch. 12 pp. 498 – 526	Studio: W, P8.1 & 8.2 (HJ 2 nd Ed).
Wk 13	4/13	Lecture: Tu, Memories (<i>TG Logic</i>); F, Memories (<i>SDRAM, Peripherals</i>)	HJS 3 rd Ed: Ch. 7 pp. 309 – 333, Ch. 8 pp. 359 – 368, 377 – 394	Studio: W, P7.5--7.8.
Wk 14	4/20	Lecture: Tu, F (<i>D/A & A/D Converters</i>)	Neamen: Ch. 16 pp. 1226 – 1235	
Wk 15	4/27	Lecture: Tu, Review; Th, F, Study Days.		FINAL EXAM DUE